

Modern Quantum Mechanics J.J. Sakurai
Solution Manual

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Chapter 1

Quantum Dynamics

1.1 Problem 2.1

$$H = -\frac{eB}{mc}S_z$$

$$\begin{aligned}\frac{dA}{dt} &= \frac{1}{i\hbar}[A, H] + \frac{\partial A}{\partial t} \\ \frac{dS_x}{dt} &= \frac{1}{i\hbar}[S_x, \omega S_z] \\ &= \frac{\omega}{i\hbar}[S_x, S_z] \\ &= \frac{\omega}{i\hbar}i\hbar(-S_y)\end{aligned}$$

$$\dot{S}_x = \omega S_y \quad (1.1)$$

Similarly,

$$\frac{dS_y}{dt} = \frac{1}{i\hbar}\omega[S_y, S_z]$$

$$\dot{S}_y = \omega S_x \quad (1.2)$$

Now, $\ddot{S}_x = -\omega\dot{S}_y = -\omega(\omega S_x)$

$$\ddot{S}_x + \omega^2 S_x = 0 \quad (1.3)$$

$$S_x(t) = A \cos \omega t + B \sin \omega t \quad (1.4)$$

$\ddot{S}_y = \omega\dot{S}_x = \omega(-\omega S_y)$

$$\ddot{S}_y + \omega^2 S_y = 0 \quad (1.5)$$

$$S_y(t) = C \cos \omega t + D \sin \omega t \quad (1.6)$$

using initial conditions at $t = 0$

$$S_x(0) = S_x \quad S_y(0) = S_y$$

we can write,

$$S_x(t) = S_x \cos \omega t + B \sin \omega t \quad (1.7)$$

$$S_y(t) = S_y \cos \omega t + D \sin \omega t \quad (1.8)$$

But from (1.1)

$$-\omega S_x \sin \omega t + \omega B \cos \omega t = -\omega S_y \cos \omega t - \omega D \sin \omega t$$

$$B = -S_y \quad D = S_x$$

$$\boxed{S_x(t) = S_x \cos \omega t - S_y \sin \omega t} \quad (1.9)$$

$$\boxed{S_y(t) = S_y \cos \omega t + S_x \sin \omega t} \quad (1.10)$$

$$\frac{dS_z}{dt} = \frac{\omega}{i\hbar} [S_z, S_z] = 0$$

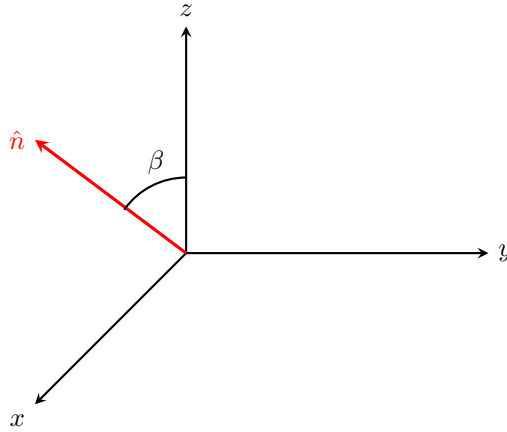
$$S_z(t) = C$$

at $t=0$ $S_z(0) = S_z$

$$\boxed{S_z(t) = S_z} \quad (1.11)$$

1.2 Problem 2.3

(a)



Choose a unit vector

$$\hat{n} = \sin \beta \hat{x} + \cos \beta \hat{z} \quad (1.12)$$

So that,

$$\begin{aligned} \vec{S} \cdot \hat{n} &= \frac{\hbar}{2} \sin \beta \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \frac{\hbar}{2} \cos \beta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ \vec{S} \cdot \hat{n} &= \frac{\hbar}{2} \begin{pmatrix} \cos \beta & \sin \beta \\ \sin \beta & -\cos \beta \end{pmatrix} \end{aligned} \quad (1.13)$$

The eigenvalues are $\pm \frac{\hbar}{2}$

Eigenvector corresponding to $\frac{\hbar}{2}$ is $\begin{pmatrix} \cos \frac{\beta}{2} \\ \sin \frac{\beta}{2} \end{pmatrix}$ therefore,

$$|\vec{S} \cdot \hat{n}; +\rangle = \cos \frac{\beta}{2} |+\rangle \sin \frac{\beta}{2} |-\rangle \quad (1.14)$$

Now, $\hat{H} = \omega S_z$

$$\begin{aligned} |\vec{S} \cdot \hat{n}; +; t\rangle &= \exp\left(-\frac{iHt}{\hbar}\right) |\vec{S} \cdot \hat{n}; +\rangle \\ &= e^{-i\frac{\omega t}{2}} \cos \frac{\beta}{2} |+\rangle + e^{i\frac{\omega t}{2}} \sin \frac{\beta}{2} |-\rangle \end{aligned} \quad (1.15)$$

Now probability of finding $s_x = \hbar/2$ will be

$$\begin{aligned} & \left| \langle S_x, + | \vec{S} \cdot \hat{n}, +; t \rangle \right|^2 \\ \langle S_x, + | \vec{S} \cdot \hat{n}, +; t \rangle &= \frac{1}{\sqrt{2}} (\langle + | + \langle - |) \left(e^{-i\omega t/2} \cos \frac{\beta}{2} |+\rangle + e^{i\omega t/2} \sin \frac{\beta}{2} |-\rangle \right) \\ &= \frac{1}{\sqrt{2}} \left(e^{-i\omega t/2} \cos \frac{\beta}{2} + e^{i\omega t/2} \sin \frac{\beta}{2} \right) \\ \left| \langle S_x, + | \vec{S} \cdot \hat{n}, +; t \rangle \right|^2 &= \frac{1}{2} \left(e^{-i\omega t/2} \cos \frac{\beta}{2} + e^{i\omega t/2} \sin \frac{\beta}{2} \right) \times \\ & \quad \left(e^{i\omega t/2} \cos \frac{\beta}{2} + e^{-i\omega t/2} \sin \frac{\beta}{2} \right) \\ &= \frac{1}{2} \left[1 + \sin \frac{\beta}{2} \cos \frac{\beta}{2} (e^{i\omega t/2} + e^{-i\omega t/2}) \right] \\ \boxed{\left| \langle S_x, + | \vec{S} \cdot \hat{n}, +; t \rangle \right|^2} &= \frac{1}{2} (1 + \sin \beta \cos \omega t) \end{aligned} \quad (1.16)$$

(b)

$$\left| \vec{S} \cdot \hat{n}, +; t \right\rangle = \begin{pmatrix} e^{-i\omega t/2} \cos \frac{\beta}{2} \\ e^{i\omega t/2} \sin \frac{\beta}{2} \end{pmatrix}$$

$$\text{let } |\alpha\rangle = \left| \vec{S} \cdot \hat{n}, +; t \right\rangle$$

$$\langle S_x \rangle = \langle \alpha | S_x | \alpha \rangle$$

$$\begin{aligned} S_x |\alpha\rangle &= \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} e^{-i\omega t/2} \cos \frac{\beta}{2} \\ e^{i\omega t/2} \sin \frac{\beta}{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \begin{pmatrix} e^{i\omega t/2} \sin \frac{\beta}{2} \\ e^{-i\omega t/2} \cos \frac{\beta}{2} \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \langle S_x \rangle &= \frac{\hbar}{2} \begin{pmatrix} e^{i\omega t/2} \cos \frac{\beta}{2} & e^{-i\omega t/2} \sin \frac{\beta}{2} \end{pmatrix} \begin{pmatrix} e^{i\omega t/2} \sin \frac{\beta}{2} \\ e^{-i\omega t/2} \cos \frac{\beta}{2} \end{pmatrix} \\ &= \frac{\hbar}{2} \cos \frac{\beta}{2} \sin \frac{\beta}{2} (e^{i\omega t} + e^{-i\omega t}) \end{aligned}$$

$$\boxed{\langle S_x \rangle = \frac{\hbar}{2} \sin \beta \cos \omega t} \quad (1.17)$$

(c)

(i)

$$\text{As } \beta \rightarrow 0 \quad \langle S_x \rangle \rightarrow 0$$

(ii)

$$\text{As } \beta \rightarrow \frac{\pi}{2} \quad \langle S_x \rangle \rightarrow \cos \omega t$$

As Expected

1.3 Problem 2.4

$$|\nu_e\rangle = \cos \theta |\nu_1\rangle - \sin \theta |\nu_2\rangle \quad (1.18)$$

$$U(t, 0) |\nu_e\rangle = |\nu_e, t\rangle$$

$$|\nu_e, t\rangle = e^{-iE_1 t/\hbar} \cos \theta |\nu_1\rangle - e^{-iE_2 t/\hbar} \sin \theta |\nu_2\rangle$$

$$\langle \nu_e | \nu_e; t \rangle = e^{-iE_1 t/\hbar} \cos^2 \theta - e^{-iE_2 t/\hbar} \sin^2 \theta$$

$$\begin{aligned} |\langle \nu_e | \nu_e; t \rangle|^2 &= \left(e^{iE_1 t/\hbar} \cos \theta \langle \nu_1 | - e^{iE_2 t/\hbar} \sin \theta \langle \nu_2 | \right) \times \\ &\quad \left(e^{-iE_1 t/\hbar} \cos \theta |\nu_1\rangle - e^{-iE_2 t/\hbar} \sin \theta |\nu_2\rangle \right) \end{aligned}$$

1.4 Problem 2.5

$$\begin{aligned}\frac{d}{dt}x(t) &= \frac{1}{i\hbar} \left[x, \frac{p^2}{2m} \right] \\ &= \frac{1}{2m} [x, p] = \frac{p}{m}\end{aligned}$$

therefore,

$$x(t) = x(0) + \frac{pt}{m} \quad (1.19)$$

$$\begin{aligned}[x(t), x(0)] &= [x(0), x(0)] + \left[\frac{pt}{m}, x(0) \right] \\ &= -\frac{t}{m} [x, p]\end{aligned}$$

$$\boxed{[x(t), x(0)] = -\frac{i\hbar t}{m}} \quad (1.20)$$

1.5 Problem 2.6

$$\begin{aligned}[[H, x], x] &= \left[\left[\frac{p^2}{2m} + V(x), x \right], x \right] \\ &= \frac{1}{2m} [[p^2, x], x] \\ &= \frac{1}{2m} [-i\hbar 2p, x] \\ &= \frac{i\hbar}{m} [x, p]\end{aligned}$$

$$[[H, x], x] = -\frac{\hbar^2}{m} \quad (1.21)$$

$$\begin{aligned}
& \sum_{a'} |\langle a'' | x | a' \rangle|^2 (E_{a'} - E_{a''}) \\
&= \sum_{a'} \langle a'' | x | a' \rangle \langle a' | x | a'' \rangle (E_{a'} - E_{a''}) \\
&= \sum_{a'} E_{a'} \langle a'' | x | a' \rangle \langle a' | x | a'' \rangle - \sum_{a'} E_{a''} \langle a'' | x | a' \rangle \langle a' | x | a'' \rangle \\
&= \sum_{a'} \langle a'' | x H | a' \rangle \langle a' | x | a'' \rangle - E_{a''} \langle a'' | x^2 | a'' \rangle \\
&= \langle a'' | x H x | a'' \rangle - \frac{1}{2} (\langle a'' | x^2 H | a'' \rangle + \langle a'' | H x^2 | a'' \rangle) \\
&= -\frac{1}{2} (\langle a'' | x^2 H + H x^2 - 2x H x | a'' \rangle) \\
&= -\frac{1}{2} \langle a'' | [[H, x], x] | a'' \rangle \\
&= -\frac{1}{2} \left(-\frac{\hbar^2}{m} \right)
\end{aligned}$$

$$\boxed{\sum_{a'} |\langle a'' | x | a' \rangle|^2 (E_{a'} - E_{a''}) = \frac{\hbar^2}{2m}} \quad (1.22)$$

1.6 Problem 2.7

$$\begin{aligned}
[\vec{x} \cdot \vec{p}, H] &= \vec{x} \cdot [\vec{p}, H] + [\vec{x}, H] \cdot \vec{p} \\
&= \vec{x} \cdot \left[\vec{p}, \frac{p^2}{2m} + V(x) \right] + \left[\vec{x}, \frac{p^2}{2m} + V(x) \right] \cdot \vec{p} \\
&= \vec{x} \cdot [\vec{p}, V(x)] + \frac{1}{2m} [\vec{x}, p^2] \cdot \vec{p} \\
&= i\hbar \vec{x} \cdot (-\nabla V) + \frac{2i\hbar}{2m} \vec{p} \cdot \vec{p}
\end{aligned}$$

$$[\vec{x} \cdot \vec{p}, H] = i\hbar \left(\frac{p^2}{2m} - \vec{x} \cdot \nabla V \right) \quad (1.23)$$

$$\begin{aligned}
\frac{d}{dt} \langle \vec{x} \cdot \vec{p} \rangle &= \frac{1}{i\hbar} \langle [\vec{x} \cdot \vec{p}, H] \rangle + \frac{\partial}{\partial t} \langle \vec{x} \cdot \vec{p} \rangle \\
&= \left\langle \frac{p^2}{2m} - \vec{x} \cdot \nabla V \right\rangle
\end{aligned}$$

$$\boxed{\frac{d}{dt} \langle \vec{x} \cdot \vec{p} \rangle = \left\langle \frac{p^2}{2m} \right\rangle - \left\langle \vec{x} \cdot \nabla V \right\rangle} \quad (1.24)$$

1.7 Problem 2.18

Ground state of 1-D Simple Harmonic Oscillator is given by

$$\psi_0(x, t) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} \exp\left(-\frac{m\omega x^2}{2\hbar} - i\frac{E_0 t}{\hbar}\right) \quad (1.25)$$

with $E_0 = \frac{\hbar\omega}{2}$. Now the correlation function

$$\begin{aligned} C(t) &= \langle x(t)x(0) \rangle \\ &= \int_{-\infty}^{\infty} \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{2}} \exp\left(\frac{m\omega x^2}{\hbar} + i\frac{\omega t}{2}\right) dx \\ &= e^{\frac{i\omega t}{2}} \end{aligned}$$

$$\boxed{C(t) = e^{\frac{i\omega t}{2}}} \quad (1.26)$$